

CSE 3102 - Lab 08: Database Exercise (2 Hours)

Database Name: MovieAnalyticsDB

Step 1: Create the following tables with appropriate constraints:

1. Movies

- movie_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- title (VARCHAR(100), NOT NULL)
- release_year (YEAR, NOT NULL)
- genre (VARCHAR(50), NOT NULL)

2. Actors

- actor_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- actor_name (VARCHAR(100), NOT NULL)
- nationality (VARCHAR(50))
- date_of_birth (DATE, NOT NULL)

3. Roles

- role_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- actor_id (INT, FK, NOT NULL)
- movie_id (INT, FK, NOT NULL)
- character_name (VARCHAR(100))
- screen_time_minutes (INT DEFAULT 0)

Step 2: Insert the following sample data:

Movies

title	release_year	genre
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Inception	2010	Sci-Fi
The Dark Knight	2008	Action
Interstellar	2014	Sci-Fi
Titanic	1997	Romance
Joker	2019	Drama

Actors

actor_name	nationality	date_of_birth
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Leonardo DiCaprio	American	1974-11-11
Christian Bale	British	1974-01-30
Anne Hathaway	American	1982-11-12
Kate Winslet	British	1975-10-05
Joaquin Phoenix	American	1974-10-28

Roles

actor_id	movie_id	character_name	screen_time_minutes
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1	1	Dom Cobb	90
1	4	Jack Dawson	80
2	2	Bruce Wayne	110
3	3	Brand	75
4	4	Rose DeWitt Bukater	85

Step 3: Write queries to solve the following:

1. Display all movies.
2. List all actors born after 1980.
3. Find all roles with screen time more than 100 minutes.
4. Count the number of movies in each genre.
5. Show each actor with total screen time.
6. Find actors who acted in more than one movie.
7. Find movies where Leonardo DiCaprio acted.
8. Show average screen time per movie.
9. List top 3 actors with highest total screen time.
10. List genres along with number of unique actors participated.